

THE RATIONALIZABILITY OF SURVEY RESPONSES

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ABSTRACT. We propose the concept of survey rationalizability, based on classical item response theory in psychology. It asks whether questions can be positioned on a common scale so that the questions endorsed by each respondent form an interval. Using ideas from classical revealed preference analysis, we characterize when survey responses are rationalizable. We then show that these results extend to polytomous questions and stochastic responses. Furthermore, we study the identification of the order of questions and respondents' ideal views, and discuss goodness-of-fit measures for surveys that are not exactly rationalizable.

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1. INTRODUCTION

Surveys are becoming increasingly popular in economics as tools for gathering data on various economic variables, such as subjective expectations, happiness, contingent valuations, beliefs, and political attitudes. Yet, the rational foundations and empirical content of survey responses remain insufficiently understood from an economic perspective. This paper takes a first step in this direction by formalizing classical psychological theories of item response and then studying their empirical content using concepts inspired by traditional revealed preference analysis. This type of exercise is crucial for assessing whether survey responses can be interpreted as measurements of a common underlying attitude and for improving surveys in terms of informativeness and accuracy.

We study surveys measuring attitudes, opinions, beliefs, or preferences on a given topic.¹ We begin the analysis with the case of dichotomous (yes/no) and deterministic responses, and later argue that this case provides the foundation for studying more complex surveys. We draw on the classical treatment of item response theory with ideal points in psychology.² Accordingly, we model rationalizability by placing survey questions on a common scale, shared by all individuals, and assigning each individual a strictly quasi-concave utility over that scale. Then, each individual endorses exactly the questions whose positions give utility above her threshold, which, by strict quasi-concavity, form an individual-specific interval.³

In Subsection 3.2, we show that the study of rationalizability can be built on the basis of an intuitive revelation principle. Consider an individual who endorses questions q_2 and q_3 , but not question q_1 . Given the interval nature of responses, the individual reveals that q_1 must occupy an extreme position in the triple. A minimal rationalizability requirement should be that these direct revelations cannot be contradictory; if

¹For instance, respondents may be asked to endorse statements on capital punishment or report participation in risky activities; see Section 3.1 for further examples.

²See Thurstone (1928) and the unfolding theory of Coombs (1964) for early developments, and see Tay and Ng (2018) and van der Linden (2018) for more recent accounts.

³We note here that our formalization of survey rationalizability can be applied to a wide variety of economic settings that deviate from the traditional framework involving choices among alternatives. The common characteristic of these settings is the absence of rivalry among the alternatives. For instance, in the context of surveys, individuals express their degree of conformity with a statement rather than choosing which of several statements they prefer. We discuss this point in greater detail in Section 3.1.

there are individuals who reveal two questions as extreme within a triple, it cannot be the case that another individual reveals the third question as also extreme in the triple. We refer to this asymmetry-type property as the Weak Axiom of Revealed Extremeness (WARE). We show that direct revelations are not sufficient for rationalizability. Subsection 3.3 then shows how to incorporate indirect revelations in two ways. First, we expand the survey data by adding particular fictitious responses so that WARE becomes sufficient for rationalizability. Second, we parallel the standard revealed preference analysis by appropriately extending the revealed extremeness relation, and then imposing asymmetry on the extended relation, a property that we call the Strong Axiom of Revealed Extremeness (SARE). In both cases, we are accounting for the exact indirect revelations contained in the data, leading to our characterization of survey rationalizability in the first part of Theorem 1. Moreover, this characterization also helps to show that testing for rationalizability is computationally tractable, as we show in the second part of the theorem. Importantly, notice that the existence of a common order is an empirical restriction rather than a property assumed to hold by construction, and one that no individual respondent could confirm or refute on her own: each response reveals only a fragment of the order, and the empirical content lies in the requirement that the fragments of disparate respondents fit together.

The analysis has two complementary uses. First, the study of rationalizability determines the order of questions compatible with the data. This may be particularly relevant when the survey designer does not know the order of questions or prefers to remain agnostic about it. This is the case in most of the settings discussed in Section 3.1 (rankings of risky activities, ideological positions extracted from text, approval voting, wishlists). But even when a candidate order is available, the rationalizability analysis can help to confirm its validity. Second, our characterization of rationalizability can also be used to assess the structural assumptions of the model. In many situations, and for various reasons, some of which are discussed below, survey responses may fail the exact rationalizability test. In this case, the same characterization provides a basis for measuring how far these responses are from rationalizability. Subsection 3.4 takes up this measurement exercise, following two standard approaches in the revealed preference literature: identifying the largest set of respondents that is rationalizable, and counting the triples on which revealed extremeness is contradicted.

We then show in Subsections 4.1 and 4.2 that our techniques for the dichotomous and deterministic case prove useful in the study of more general settings. In particular,

we consider the cases of polytomous surveys, where questions can be answered using one of multiple levels of endorsement, and probabilistic survey data, which allows for belief expressions or noisy responses. Accordingly, we adjust our notion of survey rationalizability to consider ordered collections of thresholds (one for each endorsement level) and random thresholds. Theorems 2 and 3 establish the corresponding characterizations, with both results following from simple transformations of these richer datasets into analytically equivalent dichotomous datasets, which allow us to use the techniques developed for the proof of Theorem 1.

We then shift our focus to studying the identification of these models. Since they are essentially ordinal, the key step in the analysis is determining the order of questions and where individuals may reach maximal utility. Subsection 5.1 studies the order of questions. When the survey is rationalizable, the extremeness revelations are key to learning this order. In Proposition 1, we show that a condition on the variety of responses, which we call *Richness*, guarantees that the order is unique up to inversion. Subsection 5.2 then considers the utilities of individuals. Proposition 2 demonstrates that it is possible to establish bounds for their maximizers, with the most precise information being conveyed by probabilistic surveys, whereas polytomous surveys provide more information than dichotomous surveys. Finally, Subsection 5.3 considers a specific parametric model, under which Proposition 3 shows that the locations of questions and utility maximizers can be cardinally and uniquely identified.

In Section 6 we study the unidimensional assumption. This assumption is relevant because a large number of theoretical settings and empirical applications in psychology, sociology, political science, and economics adhere to it. It is also the source of all the empirical content studied in this paper: in a multidimensional model, survey rationalizability loses its bite, since we show in Proposition 4 that every survey can be rationalized when just two dimensions are considered. Unidimensionality is thus not a simplifying assumption adopted for tractability, but the very restriction that makes the exercise testable.

Section 7 concludes by discussing several potential avenues for future research.

2. RELATED LITERATURE

In economics, the empirical literature using surveys is large and rapidly growing. Here, our focus is on its theoretical and methodological counterpart. Bertrand and

Mullainathan (2001) discuss the econometric implications of cognitive and other response errors in subjective survey data. Prelec (2004) proposes a scoring method for the elicitation of truthful subjective judgments based on answers that are more common than collectively predicted. Falk, Henkel, Neuber, and Strack (2025) develop a model in which survey responses imperfectly reveal respondents’ characteristics because of limited self-knowledge, inattention, or lack of engagement, and show how response patterns can be used to infer the precision of respondents’ signals. Manski (2004) and Stantcheva (2023) discuss several methodological issues in designing and interpreting survey studies. Benjamin, Debnam Guzman, Fleurbaey, Heffetz, and Kimball (2023) use follow-up introspective questions to study what self-reported well-being responses measure. Liu and Netzer (2023) study the use of response time to enhance the informational content of happiness survey responses, aiming to overcome the identification problems established in Bond and Lang (2019). Our contribution to this literature is to offer a theoretical framework that studies the rationality of survey responses based on classical accounts in psychology. We also show what survey responses reveal about the order of questions and the locations of individuals’ utility maximizers. Finally, our probabilistic models of survey responses enable systematic consideration of various types of errors.

Our results relate to various strands of literature across scientific disciplines, such as the use of sortability and seriation in archaeology, anthropology, biology, computer science, and psychology. The common connection among these fields is the so-called consecutive-ones property (c1p) of a 0-1 matrix. A matrix satisfies c1p if there exists a permutation of its columns such that the ones in every row are consecutive.⁴ In essence, for the dichotomous and deterministic case, survey rationalizability can be expressed as a c1p problem, enabling the use of existing classical results. In particular, we build on Tucker (1972), whose Theorem 6 supplies the implication from the absence of asteroidal triples to c1p. Our contribution is to derive the relevant ternary relation of extremeness from survey responses, to construct its indirect revelations, and to

⁴As an illustration of an application of c1p in a different field, consider the statistical archaeology problem discussed in Kendall (1969). The data consist of a set of graves and a set of objects that are either present or absent in these graves. c1p is used to determine whether graves and objects can be arranged on a time scale such that each grave contains only objects that were created and not yet obsolete during the relevant period. See Coombs and Smith (1973) and Hubert (1974) for connections between these techniques and survey responses, and Liiv (2010) for an overview of their applications across different disciplines.

show that the asymmetry of these revelations rules out the classical obstructions and delivers an economic rationalization. In doing so, we obtain a conceptually simple and tractable characterization of survey rationalizability in terms of revealed extremeness. Additionally, we show how our techniques can be extended to address more complex settings, such as those involving polytomous surveys or stochastic responses.

Our paper also relates to the study of ternary relations. Huntington and Kline (1917) and Fishburn (1971) provide early treatments of abstract ternary relations that seek to capture the notion of intermediateness on the real line. Most of our results are presented in terms of an extremeness relation, but they can be equivalently presented in terms of an intermediateness relation. Our results contribute to this literature both conceptually and technically. Conceptually, we apply ternary relations to a concrete economic problem and, instead of taking them as given, we derive them from data. Technically, we characterize when the incomplete ternary relation revealed by survey responses can be extended to the extremeness relation induced by a linear order. We achieve this by combining techniques from the c1p literature and the revealed preference literature.

3. THE RATIONALIZABILITY OF DICHOTOMOUS SURVEYS

Let $\mathcal{Q} = \{1, \dots, q, \dots, Q\}$ be a set of questions related to a specific topic, such as capital punishment or immigration, on which individuals $\mathcal{N} = \{1, \dots, n, \dots, N\}$ are surveyed. We interpret question $q \in \mathcal{Q}$ as a request to individuals to provide a graded evaluation or endorsement of the statement described in the question; in the dichotomous case considered here, these evaluations are yes/no, agree/disagree or positive/negative evaluations. Accordingly, a dichotomous survey (D-survey) is a collection of the form $\{Q_n\}_{n \in \mathcal{N}}$, with $Q_n \subseteq \mathcal{Q}$ describing the set of questions that are endorsed by individual n .⁵ Without loss of generality, we assume that all response sets are distinct and that no two questions have the same set of endorsing individuals. We study when a D-survey can be rationalized in the following sense.

D-survey rationalizability. We say that $\{Q_n\}_{n \in \mathcal{N}}$ is rationalizable whenever there exist $[\{U_n, \tau_n\}_{n \in \mathcal{N}}, \{\mu_q\}_{q \in \mathcal{Q}}]$, with $U_n : \mathbb{R} \rightarrow \mathbb{R}$ being strictly quasi-concave and $\tau_n, \mu_q \in \mathbb{R}$, such that for every $n \in \mathcal{N}$ and $q \in \mathcal{Q}$, it holds that $q \in Q_n$ if and only if $U_n(\mu_q) \geq \tau_n$.

⁵Note that by considering the incidence matrix of survey responses, we can view a D-survey as a matrix of 0s and 1s.

The notion of rationalizability contains two major assumptions: (i) views are to be located on the real line, and (ii) utility functions are strictly quasi-concave. As for the first assumption, unidimensionality is standard in psychology, sociology, political science, and many economic settings of interest. It is also what gives rationalizability its empirical content, as we discuss in Section 6. As for the second assumption, in the dichotomous and deterministic case the utility-and-threshold formulation is equivalent to the existence of a common linear order of questions under which every response set is an interval. We nevertheless retain it throughout, for two reasons. First, it gives endorsement an economic reading: an individual endorses a statement when the value she attaches to it clears a reservation level. Second, it proves useful in the study of the extensions of Section 4: one can preserve the utility function and replace the single threshold with an ordered collection of thresholds, delivering responses to polytomous surveys, or with a random threshold, delivering responses to stochastic surveys.

Before continuing, note that the interpretation of the utility function depends on the application at hand. Some surveys contain questions for which the utility function can be interpreted in classical economic terms, even though the question may be hypothetical and utility may result from the hypothetical contemplation of the question. In these situations, the threshold level may represent some internal expectation. In other types of surveys, the utility function may be interpreted as the internal value derived from supporting a specific statement, with rejected statements being those that entail high costs of departing from personal beliefs or views. The specific interpretation is not crucial to our study; the representation we adopt helps to formalize the idea of survey rationalizability in a language common to economists.

3.1. Examples of the setting. Our formal setting uses the language of attitude surveys on a given topic, which represents its most direct application. These surveys are extensively used in economics to gather information on a variety of topics and, similarly, widely used in other disciplines, such as psychology, to measure personality traits or attitudes. Examples of their use can be seen in Andrich (1988), where individuals are asked to endorse or reject a number of statements on capital punishment, and Grable and Lytton (1999), who develop a survey instrument to assess individuals' financial risk tolerance.⁶ However, our setting also applies beyond surveys. In all the examples

⁶The former includes statements/questions such as (i) capital punishment is justified because it does act as a deterrent to crime; (ii) I think capital punishment is necessary but I wish it were not; (iii) until we find a more civilized way to prevent crime we must have capital punishment. The latter

below, the data record a separate positive or negative response to each option rather than a choice among mutually exclusive options.⁷ As discussed in the Introduction, in most of the examples below the common order is not clear *ex ante*, or the analyst may prefer to remain agnostic about it.

Activities. Behavioral traits, such as risk aversion, can be analyzed by gathering information on individual participation in various potentially risky activities, such as drug use or extreme sports. This approach has a long tradition in psychology; an example is Fromme, Katz, and Rivet (1997), who study participation in risky activities.⁸ Similarly, altruism can be assessed by collecting data on participation in pro-social activities. In this type of scenario, $\mathcal{N}$ is the set of individuals surveyed about their participation in a number of activities $\mathcal{Q}$.

Text analysis. Consider the particular use of language by various individuals or institutions on a given dimension, such as behavioral traits or political attitudes. The purpose is to order a set of linguistic expressions and, as a result, to categorize individuals by their language use. See Laver, Benoit, and Garry (2003) for an application to political texts to extract policy positions.⁹ Using our terminology, Q_n represents the subset of linguistic expressions used by individual n .

Product space and recommendations. Many recommendation systems, such as those for movies or music bands, are based on arranging products and consumers within a characteristic space, with consumers favoring products closer to them.¹⁰ In some cases, objects may be arranged on a single scale, and understanding this can be instrumental in improving recommendation systems. This approach is rooted in classical psychology literature, where tastes are investigated by observing the acceptability or rejection of stimuli that can be ordered. See Pfaffmann (1960) for an early treatment, and consider,

includes questions about (i) whether it is hard to pass up a bargain; (ii) whether protection from inflation is more important than maintaining the safety of one’s money from loss or theft; and (iii) one’s comfort in investing in stocks or stock mutual funds.

⁷We use the dichotomous case, but the more general cases with Likert scales and probabilistic responses also apply to most of them.

⁸The authors gathered information on activities such as substance use (e.g., alcohol and marijuana), interpersonal behaviors (e.g., going on a blind date), and sports (e.g., rock climbing).

⁹They consider words such as ‘drugs,’ ‘markets,’ ‘nation,’ ‘secure,’ or ‘representation,’ and assign them scores on an ideological scale. See also the recent use of large language models to position political texts in several unidimensional ideological and policy spaces in Le Mens and Gallego (2025).

¹⁰The classical Hotelling setting is also based on this simple idea to ultimately model choice behavior.

as discussed there, products with different salt concentrations as an example of ordered stimuli. Following this line of thought, Q_n represents the set of products liked by individual n .

Wishlisting. The previous example involves recommendations to other users, but one may also consider recommendations to oneself in the form of a wishlist. In this context, the wishlist Q_n represents the set of items for which an individual expresses interest or a positive attitude. See Manzini, Mariotti, and Ülkü (2024) for a sequential model of individual “wishlisting.”

Attention studies. Another case that deviates from the traditional question-response paradigm of surveys involves the use of more implicit expressions of interest. This is common in attention studies, where a number of individuals $\mathcal{N}$ are presented with a number of stimuli $\mathcal{Q}$, and the analyst’s goal is to infer which stimuli attract each individual’s attention. Often, objects can be ordered, and individuals pay attention to an interval of stimuli. A classical paper in this field is Thomas (1973), which studies infants’ visual orientation toward different stimuli, including facelike stimuli.¹¹

Approval voting. The idea of allowing individuals to pick any number of options has been suggested as a voting mechanism. See Brams and Fishburn (1978) for an early analysis of approval voting. Here, Q_n represents the subset of candidates approved by individual n . Interestingly, Laslier (2009) and Alós-Ferrer and Buckenmaier (2019) have shown that equilibrium behavior in approval voting games may have the upper-contour-set structure analyzed in this paper.¹²

Judgment aggregation. There is a set of individuals $\mathcal{N}$ making judgments on a set of logical propositions $\mathcal{Q}$. Dietrich and List (2010) study conditions that facilitate judgment aggregation; one of these conditions involves the case where propositions can be ordered on a line such that the accepted judgments of each individual form an interval.

Networks and homophily. Consider a set of individuals who have directed links to others. The question is whether the network can be explained by ordering individuals such that they form links with those most similar to them. For an application using

¹¹Eye-tracking techniques are used in economics to gather information on attention; see Lahey and Oxley (2016) for a review. Polonio, Di Guida and Coricelli (2015) employ seriation techniques to understand the linkage between saccades and strategies of play within a game.

¹²Núñez and Xefteris (2017) study approval mechanisms under single-peaked preferences, which is related to our notion of endorsement-by-proximity. They show that essentially every Nash-implementable welfare optimum can be implemented by means of a generalized approval mechanism.

Twitter data to estimate political ideology, see Barberá (2015). We can understand this within our setting by considering $\mathcal{Q} = \mathcal{N}$.

Matching. Let $\mathcal{N}$ be a set of individuals and $\mathcal{Q}$ a set of objects (as in housing problems) or a set of different individuals (as in marriage problems). Individuals submit a set Q_n of acceptable elements. Typically, individuals have a common preference ordering over $\mathcal{Q}$ determined, e.g., by quality. However, in some settings, individuals may have idiosyncratic single-peaked preferences on $\mathcal{Q}$, perhaps given by quasi-linear preferences on quality and price in the housing problem, or by affinity in the marriage problem (see Bade (2019)).

3.2. Revealed information: extremeness. We start by discussing how survey responses reveal the order of questions, which will soon be proven to be the critical feature for survey rationalizability. First, recall that the upper contour set of a strictly quasi-concave utility function must correspond to an interval on the real line. Then, consider individual n , and suppose we observe $Q_n \cap \{q_1, q_2, q_3\} = \{q_2, q_3\}$. In other words, individual n endorses questions q_2 and q_3 but not question q_1 . Given the interval nature of responses, this observation reveals that q_1 occupies an extreme position within the triple. When this is the case, we write $q_1 \mid \{q_2, q_3\}$ and say that q_1 has been revealed as extreme in the triple $\{q_1, q_2, q_3\}$.

Each respondent generally reveals only part of the order, so a common-order rationalization requires the individual revelations to be mutually consistent. This motivates the following asymmetry property.

Weak Axiom of Revealed Extremeness (WARE). We say that $\{Q_n\}_{n \in \mathcal{N}}$ satisfies WARE if, for every triple $\{q_1, q_2, q_3\}$ of questions, whenever $q_1 \mid \{q_2, q_3\}$ and $q_2 \mid \{q_1, q_3\}$ hold, $q_3 \mid \{q_1, q_2\}$ does not hold.

Once $q_1 \mid \{q_2, q_3\}$ and $q_2 \mid \{q_1, q_3\}$ have been revealed, rationalization requires positioning q_1 and q_2 as extremes within the triple, which in turn necessitates that q_3 is not extreme. It is immediate that WARE is a necessary property for rationalizability. To see this, suppose that $\{Q_n\}_{n \in \mathcal{N}}$ is rationalizable and assume that $q_1 \mid \{q_2, q_3\}$ and $q_2 \mid \{q_1, q_3\}$ hold. Then, it must be that $\{\mu_{q_1}, \mu_{q_2}\} = \{\min\{\mu_{q_1}, \mu_{q_2}, \mu_{q_3}\}, \max\{\mu_{q_1}, \mu_{q_2}, \mu_{q_3}\}\}$. Hence, if we observe any (interval) response set containing both q_1 and q_2 , it must also contain q_3 , and WARE is satisfied. However, the following example shows that WARE is not sufficient for rationalization.

Example 1. Consider $\mathcal{N} = \{1, 2, 3\}$ and $\mathcal{Q} = \{q_1, q_2, q_3, q_4\}$, with response sets $Q_1 = \{q_1, q_2\}$, $Q_2 = \{q_1, q_3\}$, and $Q_3 = \{q_1, q_4\}$. By construction, there are at most two extremeness revelations in every triple of alternatives, and WARE holds. However, the responses are not rationalizable; the information contained in Q_1 requires q_4 to occupy either the highest or lowest position in the triple $\{q_1, q_2, q_4\}$. If it is a maximum (respectively, a minimum), the information contained in Q_2 requires q_4 to also be the maximum (respectively, the minimum) in the triple $\{q_1, q_3, q_4\}$ and, as a consequence, the maximum (respectively, the minimum) in the set $\{q_1, q_2, q_3, q_4\}$. The same argument places each of q_3 and q_2 at an extreme of the four-question set. A linear order has only two extreme questions, yielding a contradiction.

3.3. A characterization of rationalizability. Example 1 shows that WARE is not sufficient because it only rules out direct contradictions. We now present two equivalent ways of incorporating indirect information. The first augments the survey with fictitious responses; the second extends the extremeness relation to include all indirect revelations. These approaches lead to our characterization and are also useful for studying computational complexity.

In order to discuss the first approach, we need to consider pairs of overlapping responses, namely:

$$(Q_i, Q_j) \text{ such that } Q_i \cap Q_j \neq \emptyset, Q_i \setminus Q_j \neq \emptyset \text{ and } Q_j \setminus Q_i \neq \emptyset.$$

Recall that, under rationalizability, response sets correspond to intervals of the real line. Hence, the union of two overlapping response sets also corresponds to a (larger) interval.¹³ If this union is not already a response set in the original survey, it brings new information that can be added to the original survey. We then consider this principle recursively until no more novel unions of overlapping responses can be added. Denote $\mathcal{N}_0 = \mathcal{N}$ and consider a sequence of D-surveys $\{Q_n\}_{n \in \mathcal{N}_m}$, $m = 0, 1, \dots, M$, such that for every $m < M$, the $(m+1)$ -th D-survey is equal to the m -th D-survey with the addition of a (novel) union of two overlapping sets in the m -th D-survey. Given finiteness, this sequence terminates at a D-survey that admits no further addition,

¹³Notice that this statement does not necessarily apply if the responses are not overlapping: if two intervals are disjoint, the union may fail to be an interval. Also, notice that the intersection and difference of overlapping intervals must also be intervals, but our revelation technique does not require including this information.

$\{Q_n\}_{n \in \mathcal{N}_M}$, which we call the expanded D-survey of $\{Q_n\}_{n \in \mathcal{N}}$. Importantly, notice that this survey is closed under unions of overlapping responses.¹⁴

As for the second approach, we now define the indirect extremeness revelations derived from the direct extremeness revelations contained in the original D-survey. Let T be a generic ternary relation on a finite set X , and define the following condition, which we call overlapping:¹⁵

$$xT\{y, z_1\} \text{ and } xT\{y, z_2\} \text{ implies } xT\{z_1, z_2\}.$$

Given T , the overlapping closure of T , denoted $\widehat{T}$, is the minimal ternary relation containing T that is closed under overlapping. We then use the overlapping closure of $\mid$ to define a stronger version of WARE.

Strong Axiom of Revealed Extremeness (SARE). We say that $\{Q_n\}_{n \in \mathcal{N}}$ satisfies SARE if, for every triple $\{q_1, q_2, q_3\}$ of questions, whenever $q_1 \widehat{\mid} \{q_2, q_3\}$ and $q_2 \widehat{\mid} \{q_1, q_3\}$ hold, $q_3 \widehat{\mid} \{q_1, q_2\}$ does not hold.

We are now in a position to state the main theorem of this section, Theorem 1. It shows that D-survey rationalizability is equivalent to the expanded D-survey satisfying WARE, which is itself equivalent to the D-survey satisfying SARE. Importantly, Theorem 1 shows that checking the rationalizability of data can be done in polynomial time, making the problem computationally tractable.

Theorem 1. *The following statements are equivalent:*

- (1) $\{Q_n\}_{n \in \mathcal{N}}$ is rationalizable.
- (2) $\{Q_n\}_{n \in \mathcal{N}_M}$ satisfies WARE.
- (3) $\{Q_n\}_{n \in \mathcal{N}}$ satisfies SARE.

Moreover, determining whether $\{Q_n\}_{n \in \mathcal{N}}$ is rationalizable can be done in polynomial time, with a complexity of $O(NQ^3 + Q^4)$.

Theorem 1 demonstrates that survey rationalizability can be understood through a familiar economic approach: (i) include all indirect revelations derivable from the data, and (ii) require these revelations to be asymmetric. These revelations concern attitude survey responses rather than choices: they describe the positions of questions on the

¹⁴The order in which information is added is irrelevant and the expanded D-survey is unique.

¹⁵The property is a type of transitivity and we use the term overlapping to relate it to the notion of overlapping sets.

attitude scale, not a decision-maker’s preference ordering. They also form a ternary rather than a binary relation.

The proof of the equivalence of statements (2) and (3) is recursive: each fictitious response added generates only revelations obtainable by overlapping, and every revelation in the overlapping closure is eventually generated by a response in the expanded survey.

The equivalence of (2) and (3) with statement (1) is established as follows. First, survey rationalizability can be expressed in terms of the ordinal consecutive-ones property (c1p).¹⁶ In essence, for a given order of questions $\succ$ where c1p holds, one can construct a corresponding strictly quasi-concave utility function and threshold for each individual. This utility function is strictly increasing up to the midpoint of the interval defined by the questions with the minimum and maximum locations among those endorsed. At these two boundary questions, the utility is zero, which is equal to the threshold. The consecutive nature of responses given by c1p ensures this construction is valid for rationalizability.

In order to show that our properties guarantee c1p, we use Theorem 6 in Tucker (1972), which is based on the absence of “asteroidal triples,” a concept introduced by Lekkerkerker and Boland (1962) in their study of interval graphs. In essence, a violation of the asymmetry condition contained in WARE corresponds to the simplest possible asteroidal triple, and we need to prove that the asymmetry of indirect revelations rules out any asteroidal triple. This result follows from a recursive argument, with the indirect revelation approach facilitating the analysis.¹⁷

Finally, given the equivalence between (1) and (3), the complexity of checking for rationalizability reduces to computing the extremeness relation of the original D-survey and checking the asymmetry of its overlapping closure. The first step requires reading

¹⁶See the proof below for the formal definitions of c1p and of an asteroidal triple in our context.

¹⁷Bergman (2022) offers an alternative potential route. Using our terminology, he works with the smallest collection of response sets that contains the original response data and is closed under unions, intersections, and differences of overlapping sets. Bergman shows that a simple property on this expanded dataset characterizes the existence of a linear order on the collection of questions. For his result to be useful in our setting, one would need to connect his larger expanded dataset to a unique ternary relation and a unique transitivity operation. This does not appear to be straightforward; doing so remains an interesting direction for future research. By contrast, we only consider the union of overlapping sets, and our use of Tucker (1972) allows us to create paths between alternatives and connect these paths to our ternary relation and the simple property WARE.

the data once, in $O(NQ^3)$ operations, and is the only one in which the number of respondents intervenes. For the second step, we show in the proof that this can be done using the classical Floyd-Warshall algorithm, applied once for each question, at a cost of $O(Q^4)$ that depends on the number of questions alone.

3.4. Partial rationalizability. The discussion so far concerns exact rationalizability: the entire survey must be compatible with a single common order of questions. We view this requirement as a benchmark. As in traditional revealed preference analysis, the value of an exact characterization lies not only in providing a test, but also in providing a basis for measuring departures from rationalizability. Here, we briefly consider two complementary measures, defined at the level of respondents and revealed triples.

Respondents. The rationalizable fraction of respondents is

$$\rho_R = \max \left\{ \frac{|\mathcal{M}|}{N} : \mathcal{M} \subseteq \mathcal{N} \text{ and } \{Q_n\}_{n \in \mathcal{M}} \text{ is rationalizable} \right\}.$$

Thus, $1 - \rho_R$ is the minimal fraction of respondents that must be set aside for the remaining responses to be organized by one common scale, in the spirit of the index of Houtman and Maks (1985).¹⁸ By Theorem 1, testing a given subset of respondents is inexpensive, so the cost of computing ρ_R lies in the search over subsets. That type of search has been studied in the clp literature; see, e.g., Narayanaswamy and Subashini (2015). Applied to our survey setting, what is known there is that the problem is fixed-parameter tractable in the number of respondents removed: the running time is polynomial in the size of the survey and exponential only in $N(1 - \rho_R)$. The cost of computing ρ_R is therefore governed by the value of ρ_R itself: the closer the survey is to being rationalizable, the fewer respondents have to be set aside, and the cheaper the index is to obtain.

Revealed triples. Let $\mathcal{V}$ denote the collection of triples of distinct questions $\{q_1, q_2, q_3\}$ for which

$$q_1 \widehat{|\{q_2, q_3\}}, \quad q_2 \widehat{|\{q_1, q_3\}}, \quad \text{and} \quad q_3 \widehat{|\{q_1, q_2\}}.$$

For $Q \geq 3$, define

$$\rho_T = 1 - \frac{|\mathcal{V}|}{\binom{Q}{3}}.$$

¹⁸See Apesteguia and Ballester (2015) for a study of this and related indices. Note that, unlike for the rationalizability test, identical response sets must here be retained with their multiplicities.

This is the counterpart, for our ternary relation, of counting violations of a consistency property in individual choice, as proposed by Swofford and Whitney (1987) and Famulari (1995).¹⁹ The score records the frequency of triples that comply with SARE. Like ρ_R , ρ_T takes the value one exactly when the survey is rationalizable. The index ρ_T is at the heart of Theorem 1, and its computation requires nothing beyond that of Theorem 1. The algorithm behind that result already constructs the overlapping closure of the extremeness relation and inspects its triples. Computing ρ_T changes only this last operation: instead of checking whether some triple fails, one counts how many do. Its complexity is therefore that of the rationalizability test itself, $O(NQ^3 + Q^4)$. Our characterization thus delivers more than a yes/no test: the same object that decides rationalizability also measures, in a tractable way, the extent of its failure.

4. POLYTOMOUS AND STOCHASTIC SURVEYS

In this section, we show how to exploit the structure of Theorem 1 when dealing with two related, but informationally richer, problems. The first one covers the Likert scale case in which individuals may use more than two labels to declare their endorsement. The second analyzes the case of dichotomous surveys, in which responses may be probabilistic.

4.1. Polytomous surveys. Given question $q \in \mathcal{Q}$, individual $n \in \mathcal{N}$ is now allowed to express a level of endorsement by using a label from $\mathcal{L} = \{0, 1, \dots, L\}$, where $L > 0$. We write $\mathcal{L}^* = \mathcal{L} \setminus \{0\}$. Higher labels in the collection express stronger endorsement.²⁰ A polytomous survey or, simply, a P-survey is a map $P : \mathcal{N} \times \mathcal{Q} \rightarrow \mathcal{L}$, where $P(n, q) = l$ refers to the case where individual n assigns label l to question q . We consider the following notion of rationalizability, which reduces to the notion used for D-surveys when $L = 1$.

P-survey rationalizability. We say that P is rationalizable whenever there exist $[\{U_n, \{\tau_n^l\}_{l \in \mathcal{L}^*}\}_{n \in \mathcal{N}}, \{\mu_q\}_{q \in \mathcal{Q}}]$, with $U_n : \mathbb{R} \rightarrow \mathbb{R}$ being strictly quasi-concave, $\tau_n^1 < \dots < \tau_n^L$, and $\mu_q \in \mathbb{R}$ such that, for every $n \in \mathcal{N}$, $q \in \mathcal{Q}$ and $l \in \mathcal{L}^*$, $P(n, q) = L$ if $U_n(\mu_q) \geq \tau_n^L$, $P(n, q) = l$, $0 < l < L$, if $\tau_n^{l+1} > U_n(\mu_q) \geq \tau_n^l$ and $P(n, q) = 0$ otherwise.

¹⁹For consumption data, Koo (1963) is an early antecedent, and Echenique, Lee, and Shum (2011) and Dean and Martin (2016) refine the idea by weighting cycles, respectively by the wealth lost along them and by the minimal cost of the revealed preference relations that must be removed to eliminate them.

²⁰For instance, strongly disagree, disagree, neutral, agree, strongly agree.

In words, the individual now uses different thresholds to determine her degree of endorsement. These thresholds naturally impose stronger requirements on stronger expressions of endorsement.

We now show that the rationalizability of a P-survey can be obtained building upon Theorem 1, by studying the rationalizability of an associated D-survey. Formally, given the response map P , define the D-survey

$$\{Q_{(n,l)}^P\}_{(n,l) \in \mathcal{N} \times \mathcal{L}^*} \text{ where } Q_{(n,l)}^P = \{q \in \mathcal{Q} : P(n, q) \geq l\}.$$

That is, the vector of responses of individual $n \in \mathcal{N}$ is used to construct L response sets. Each of these response sets contains cumulative information, e.g., response set $Q_{(n,l)}^P$ contains all the questions endorsed by individual n with intensity l or above. Note that, formally, $\{Q_{(n,l)}^P\}_{(n,l) \in \mathcal{N} \times \mathcal{L}^*}$ is a D-survey where every (n, l) can be understood as an individual. It is immediate that $\{Q_{(n,l)}^P\}_{(n,l) \in \mathcal{N} \times \mathcal{L}^*}$ provides the same information as P does. We need to define the corresponding expanded D-survey, following the same logic as above, denoted by $\{Q_n^P\}_{n \in \mathcal{N}_M}$. Theorem 2 shows that rationalizability of P is equivalent to this expanded D-survey satisfying WARE.²¹

Theorem 2. *P is rationalizable if and only if $\{Q_n^P\}_{n \in \mathcal{N}_M}$ satisfies WARE.*

4.2. Stochastic responses. We now study the case where survey responses may be stochastic. This can be understood either as variation in the responses of individuals within a given subgroup, such as an age or gender group, or as a case in which individual responses express a value on a continuum, such as a probabilistic belief, expectation, or intensity of support. Stochastic models of responses are also instrumental in dealing with actual data, since they are a way to model and account for noise resulting from, e.g., mistakes due to fatigue or inattention.

Formally, a dichotomous survey with stochastic responses, or S-survey, is a map $S : \mathcal{N} \times \mathcal{Q} \rightarrow [0, 1]$ where $S(n, q)$ describes the probability with which individual $n \in \mathcal{N}$ endorses question $q \in \mathcal{Q}$. Consider the following definition of rationalizability.

S-survey rationalizability. We say that S is rationalizable whenever there exist $[\{U_n, F_n\}_{n \in \mathcal{N}}, \{\mu_q\}_{q \in \mathcal{Q}}]$, with $U_n : \mathbb{R} \rightarrow \mathbb{R}$ being strictly quasi-concave, $F_n : \mathbb{R} \rightarrow [0, 1]$ being a cumulative distribution function (CDF) governing the realization of thresholds τ_n , and $\mu_q \in \mathbb{R}$ such that, for every $n \in \mathcal{N}$ and $q \in \mathcal{Q}$, $S(n, q) = F_n(U_n(\mu_q))$.

²¹For brevity, this and the following result present the characterization based solely on expanded surveys. Nevertheless, the conclusions about overlapping closure and associated complexity remain equally applicable.

A question is endorsed whenever the utility of the view it represents is at least as high as the realized threshold, which is distributed according to F_n . Hence, endorsement occurs probabilistically, and the probability that individual n endorses question q is the mass of thresholds at or below $U_n(\mu_q)$, that is, $F_n(U_n(\mu_q))$. This is a convenient account of stochastic responses that aligns well with the models studied in this paper and can easily accommodate mistakes at the time of responding. For example, F_n may place high probability on a benchmark threshold and small probability on lower or higher thresholds. The individual may then occasionally endorse a question whose utility lies below the benchmark or reject one whose utility lies above it.

We now argue that we can rely again on the techniques developed for the analysis of D-surveys. For every $n \in \mathcal{N}$, consider the (at most) Q response sets defined by $Q_{(n,q)}^S = \{q' \in \mathcal{Q} : S(n, q') \geq S(n, q)\}$, and the corresponding D-survey $\{Q_{(n,q)}^S\}_{(n,q) \in \mathcal{N} \times \mathcal{Q}}$. That is, the information about individual $n \in \mathcal{N}$ is represented at most Q times, describing response sets that contain the questions chosen with at least as much probability as a given question. Denote the corresponding expanded D-survey by $\{Q_n^S\}_{n \in \mathcal{N}_M}$. Theorem 3 shows that rationalizability of S is equivalent to this expanded D-survey satisfying WARE.

Theorem 3. *S is rationalizable if and only if $\{Q_n^S\}_{n \in \mathcal{N}_M}$ satisfies WARE.*

5. IDENTIFICATION

In this section, we discuss the identification of the models studied. To present results across different types of surveys, we make the following assumptions for every individual $n \in \mathcal{N}$: (i) $\emptyset \neq Q_n \neq \mathcal{Q}$, and (ii) $S(n, q) \geq S(n, q') \Rightarrow P(n, q) \geq P(n, q') \Rightarrow [q' \in Q_n \text{ implies } q \in Q_n]$. That is, (i) implies that we eliminate from the analysis the trivial individuals that endorse everything or nothing, and (ii) connects the response behavior in S-surveys, P-surveys and D-surveys, by requiring the three formats to rank questions in a consistent manner. Throughout this section we assume that surveys are rationalizable with the locations of questions and the utility functions being $\{\mu_q\}_{q \in \mathcal{Q}}$ and $\{U_n\}_{n \in \mathcal{N}}$. We now ask what the data reveal about these objects. In so doing, we impose the following assumption throughout.

Richness. For every triple of consecutive questions $\mu_{q_1} > \mu_{q_2} > \mu_{q_3}$, the expanded D-survey contains two, possibly fictitious, individuals $n_1, n_2 \in \mathcal{N}_M$ such that $Q_{n_1} \cap$

$\{q_1, q_2, q_3\} = \{q_1, q_2\}$ and $Q_{n_2} \cap \{q_1, q_2, q_3\} = \{q_2, q_3\}$. For P-surveys and S-surveys, the condition applies to the expansion of the corresponding associated D-survey.²²

In words, Richness asks the data to separate neighboring questions: for every three consecutive questions, the expanded survey must contain a response holding the upper two and excluding the lowest, and another holding the lower two and excluding the highest. The condition is demanding, and it is demanding precisely where questions are most alike and the distinctions between them least consequential. This is the price of recovering the entire order exactly, and not of rationalizability, which Theorem 1 characterizes without any such requirement; as noted after Proposition 1, the order remains partially identified when Richness fails.

5.1. Identification of questions. As discussed above, the models are essentially ordinal, so we need to study the identification of the linearly ordered set $(Q, \succ)$ corresponding to the linearly ordered set $(\{\mu_q\}_{q \in Q}, >)$. We start by showing that $\succ$ is fully identified up to inversion; that is, we show that all possible rationalizations of the survey responses locate the questions in the order given by $\succ$ or its inverse. Under Richness, this follows because the extremeness revelations are complete.²³

Proposition 1. *Given Richness, the indirect revealed relation coincides with the extremeness relation induced by $\succ$. As a result, $\succ$ is identified up to inversion.*

Even without Richness, the same idea yields partial identification: any linear order whose extremeness relation contains the relation revealed by the expanded D-survey remains admissible.

5.2. Identification of individuals. We now discuss what can be learned about a given individual $n \in \mathcal{N}$. Whenever U_n attains a maximum, strict quasi-concavity guarantees that the maximizer is unique; we denote it by μ_n and call it the ideal view of individual n . Given the finiteness of Q , we can restrict attention to rationalizations of this type without loss of generality.²⁴

²²As will become clear, this requirement can be satisfied with $Q - 1$ individuals in the expanded survey. If every triple involves different pairs of individuals, then $2(Q - 2)$ individuals would be needed.

²³Since this result holds for all types of surveys, we do not specify the type in the formulation of the result. Also, we simplify the exposition of the proof by showing the key steps using only the D-survey notation.

²⁴Any rationalizable survey admits an observationally equivalent rationalization in which U_n attains a maximum. In particular, an individual with strictly increasing (decreasing) utility, and hence with

Since the model is ordinal and given the uniqueness established in Proposition 1, here we focus on the identification of the ordinal location of the ideal view of the individual with respect to $(\mathcal{Q}, \succ)$. Given an α -survey, with $\alpha \in \{D, P, S\}$, we call the identified set of the ideal view of individual n the collection of positions relative to $(\mathcal{Q}, \succ)$ that the ideal view of n may occupy across all possible rationalizations of the survey. When the three survey formats are related by assumption (ii), we show that S-surveys are weakly more informative in this respect than P-surveys, which in turn are weakly more informative than D-surveys.

For any individual $n \in \mathcal{N}$, denote by $\underline{q}_n^\alpha$ and $\bar{q}_n^\alpha$, with $\alpha \in \{D, P, S\}$, the minimum and maximum questions according to $\succ$ among (a) the endorsed ones when $\alpha = D$, (b) those with the highest strength of endorsement when $\alpha = P$, and (c) those with the highest probability of endorsement when $\alpha = S$. Then, denote by $\alpha(n)$ the set of positions relative to $\succ$ lying strictly between the question immediately below $\underline{q}_n^\alpha$ and the one immediately above $\bar{q}_n^\alpha$ in $\succ$.²⁵

Proposition 2. *Given Richness, the identified set of the ideal view of individual n in an α -survey is exactly $\alpha(n)$. Moreover, $S(n) \subseteq P(n) \subseteq D(n)$.*

5.3. Exponential responses. We now introduce a stochastic parametric model under which the locations of questions and ideal views, $\{\mu_q\}_{q \in \mathcal{Q}}$ and $\{\mu_n\}_{n \in \mathcal{N}}$, can be cardinally identified. The model specifies the individual's endorsement probability as an exponential function of the distance between the question and an ideal view, $S(n, q) = e^{-\frac{|\mu_n - \mu_q|}{\sigma_n}}$.²⁶ Intuitively, the probability of endorsing a question is 1 when its location coincides with that of the ideal view, decays with the distance between them, and is zero in the limit. It is modulated by a single parameter $\sigma_n > 0$ with a natural interpretation. Smaller values of σ_n accelerate the decay, representing individuals with a lower tendency to endorse views away from their ideal view. This is the ideal-point form in which models of endorsement-by-proximity are usually written in the psychology literature and is a particular case of our stochastic framework: set $U_n(x) = e^{-\frac{|\mu_n - x|}{\sigma_n}}$

no peak on the reals, is indistinguishable in ordinal terms from an individual with a peak above the maximum (below the minimum) of all questions according to $\succ$.

²⁵If one of these two questions does not exist, $\alpha(n)$ is unbounded in the corresponding direction.

²⁶The literature in psychology offers various probabilistic models of responses based on the notion of endorsement-by-proximity. See, e.g., Davison (1977). The present model is a convenient specification related to Andrich (1988) and Hoijtink (1990).

and let F_n be the CDF of the uniform distribution on $[0, 1]$. We normalize the locations of two questions, denoted by 0_q and 1_q , by setting $\mu_{0_q} = 0$ and $\mu_{1_q} = 1$, respectively.²⁷

Proposition 3. *Assume Richness and $Q \geq 4$. In an S -survey with exponential responses, $\{\mu_q\}_{q \in \mathcal{Q} \setminus \{0_q, 1_q\}}$ and $\{\mu_n\}_{n \in \mathcal{N}}$ are cardinally identified.*

6. MULTIDIMENSIONALITY

We now allow the space of question locations and the domain of utilities to be multidimensional. Given the result below, it suffices to consider D-surveys and the two-dimensional space $\mathbb{R}^2$. We now show that, without further restrictions, D-survey rationalizability in a two-dimensional space has no empirical content.²⁸

Proposition 4. *Every D-survey is rationalizable in a two-dimensional space.*

Whereas in the unidimensional case only two questions can be extreme, two-dimensional representations permit arbitrarily many. This flexibility leaves D-survey responses unrestricted. Furthermore, the same result holds even if the rationalization concept is strengthened to require strictly convex contour sets.²⁹ Thus, without further parametric restrictions, D-survey rationalizability in a multidimensional space has no empirical content.

This result is unsurprising given the existing literature on convex representations of preference profiles. Bogomolnaia and Laslier (2007) demonstrate in their Theorem 16 that every preference profile admits a convex representation in two dimensions. In the current survey setting, we work with a single upper contour set per individual rather than the entire preference profile. As a result, our Proposition 4 can be viewed as a corollary to Theorem 16 of Bogomolnaia and Laslier (2007).³⁰

²⁷Recall that the inverse of a linear order is equivalent to the linear order for our purposes, so the order of these two alternatives is without loss of generality.

²⁸D-survey rationalizability in a two-dimensional space follows the definition of D-survey rationalizability, except that the space of question locations and the domain of utilities are now $\mathbb{R}^2$. All other objects are unaffected.

²⁹Although the indifference curves used in our construction are polygons formed by a finite number of points on the circumference of the unit circle, a sufficiently small quadratic perturbation of the utility function makes it strictly concave, and hence strictly quasi-concave, without altering any response.

³⁰Notice, though, that the critical unidimensional cases are fundamentally different. Unlike other settings where information on the preference profile is assumed, such as Kalandrakis (2010) or Ballester and Haeringer (2011), the study of survey responses involves only very coarse information about individuals—specifically, a single upper contour set per individual.

Naturally, one might instead impose a parametric geometric restriction in several dimensions. Consider Euclidean preferences, under which individual n endorses precisely the questions located in a disk centered at her ideal view, with an individual-specific radius. This restriction has empirical content. To see this, consider four individuals and, for every $\mathcal{M} \subseteq \{1, 2, 3, 4\}$, a question $q_{\mathcal{M}}$ endorsed exactly by the individuals in $\mathcal{M}$. A rationalization would require four disks realizing all 16 membership patterns. Since the set of question locations is finite, the radius of each disk can be increased slightly without changing any response; hence, we may assume that no question lies on a boundary circle. The four boundary circles partition the plane into at most 14 regions, and hence the 16 patterns cannot all be realized.³¹ Characterizing rationalizability under Euclidean preferences, possibly as a function of the number of dimensions, is an interesting direction for future research.

7. DISCUSSION

We conclude by commenting on a number of possible extensions. An immediate one is to model heterogeneity in perceived orders. Subsection 3.4 measures departures from a single common order, but it does not model their source. One could instead ask for the minimal number of orders needed to organize the population, or partition respondents into groups that are homogeneous in the scale they use; different demographic groups, by age or gender, may well entertain different orders of questions. Characterizing the populations that admit a small number of perceived orders is a natural direction for future research.

Another direction of interest is the combination of surveys with more standard economic data to enhance the information revelation process, in line with the recent suggestions of Caplin (2025), Gerasimou (2021) and Almås, Attanasio, and Jervis (2024).

Also, it is interesting to note that our framework remains valid in the presence of non-truthful responses and, indeed, it provides grounds to empirically study the nature of the potentially existing deviations. Notice first that a respondent trying to appear, say, more moderate may present a response set in the form of an interval, but shifted towards the moderate part of the scale. Hence, our techniques can still be used in the presence of non-truthful responses, providing correct information on the order of questions, and on the self-presentation of the individuals, who could deviate from their internal views due

³¹See, e.g., Ruskey, Savage, and Wagon (2006) for this and related counts, which follow from Euler's relation for planar graphs.

to conformity, shame, etc. As is common in the survey literature, we could compare survey responses collected under different conditions (e.g., degrees of confidentiality or anonymity, different informational environments, or different incentives), or use implicit information (such as response times or measures of attention) to better identify individuals' actual views. The ordered nature of questions obtained in the analysis would facilitate this. Moreover, we can shed light on which variables of the model, e.g., the threshold level or the individual utilities, are affected and how.

Finally, our framework could provide the basis to study the impact of behavioral considerations on responses. As discussed, our stochastic treatment studied in Section 4.2 provides a flexible tool to incorporate considerations such as mistakes. A possible line of future research would entail explicitly modeling the nature of mistakes. For example, respondents may be affected by fatigue and, as a result, increase or decrease their threshold over the course of the survey.

APPENDIX A. PROOFS

Proof of Theorem 1: (1) $\Rightarrow$ (2). Consider any appropriate sequence of D-surveys $\{Q_n\}_{n \in \mathcal{N}_m}$, $m = 0, 1, \dots, M$, leading to the expanded survey. Given (1), $\{Q_n\}_{n \in \mathcal{N}_0}$ is rationalizable. We now claim that if $\{Q_n\}_{n \in \mathcal{N}_m}$ is rationalizable, $\{Q_n\}_{n \in \mathcal{N}_{m+1}}$ is also rationalizable. To see this, suppose that $\{Q_n\}_{n \in \mathcal{N}_m}$ is rationalizable with parameters $[\{U_n, \tau_n\}_{n \in \mathcal{N}_m}, \{\mu_q\}_{q \in \mathcal{Q}}]$. Let $Q_i \cup Q_j$ be the novel response set added to form $\{Q_n\}_{n \in \mathcal{N}_{m+1}}$. This requires Q_i, Q_j to belong to $\{Q_n\}_{n \in \mathcal{N}_m}$ and to be overlapping. To construct the new rationalization, define the location of questions and the individual parameters of any individual other than the novel one as in the rationalization of $\{Q_n\}_{n \in \mathcal{N}_m}$; we then need to define parameters for the novel individual and prove the rationalization of her response set $Q_i \cup Q_j$. To do so, define $a = \min\{\mu_q : q \in Q_i \cup Q_j\}$, $b = \max\{\mu_q : q \in Q_i \cup Q_j\}$ and define the utility of the novel individual to be any strictly increasing utility function up to $\frac{a+b}{2}$ and strictly decreasing after this point, with the utility at a and b being equal to the threshold level. Given the original rationalizability and the overlapping nature of Q_i and Q_j , the interval $[a, b]$ contains exactly the questions in $Q_i \cup Q_j$, and rationalizability of this response set follows. Given finiteness, this inductive argument guarantees that the expanded D-survey is rationalizable and, as shown in Section 3.2, it must satisfy WARE.

(2) $\Rightarrow$ (1). We use the characterization of c1p in Theorem 6 in Tucker (1972), via the absence of asteroidal triples. In order to be able to use it, we first need to bring

the concepts used there to our setting. Formally, c1p requires the existence of a linear order $\succ$ over $\mathcal{Q}$ such that every response set Q_n is formed by consecutive questions in $\succ$. An asteroidal triple is a triple of questions $\{q_1, q_2, q_3\}$ such that for every pair of questions $\{q_i, q_j\}$ in this triple, there exists a sequence of individuals $n_{ij}^1, \dots, n_{ij}^K$ such that (i) $q_i \in Q_{n_{ij}^1}$ and $q_j \in Q_{n_{ij}^K}$, (ii) for every $k = 1, \dots, K-1$, $Q_{n_{ij}^k} \cap Q_{n_{ij}^{k+1}} \neq \emptyset$, and (iii) the remaining alternative in the triple does not belong to the response set of any individual in this sequence.

We now show that if the expanded D-survey satisfies WARE, the original survey does not contain asteroidal triples. Suppose, for contradiction, that $\{q_1, q_2, q_3\}$ is an asteroidal triple. Consider first the response sets of n_{ij}^1 and n_{ij}^2 . Given (ii), it must be either that one of these sets contains the other, or that they overlap. In the former case, the larger of these two response sets, denoted $\bar{Q}_{ij}^2$, belongs to the expanded D-survey (as it belongs to the survey), contains q_i but not the remaining alternative from the triple that we denote by q_o , and intersects $Q_{n_{ij}^3}$. In the latter case, since the expanded D-survey is closed under overlapping unions, $\bar{Q}_{ij}^2 = Q_{n_{ij}^1} \cup Q_{n_{ij}^2}$ must belong to the expanded D-survey, contains q_i but not q_o , and intersects $Q_{n_{ij}^3}$. Consider the response sets $\bar{Q}_{ij}^2$ and $Q_{n_{ij}^3}$. Again, it must be either the case that one of them contains the other or they overlap. With the same reasoning, we can identify $\bar{Q}_{ij}^3$ in the expanded D-survey, containing q_i but not q_o , and intersecting $Q_{n_{ij}^4}$. Proceed recursively until obtaining the response set $\bar{Q}_{n_{ij}^K}$.³² This set contains q_i and q_j , but not q_o . Since this argument applies to any combination in the triple, we have identified three (fictitious) individuals in the expanded D-survey violating WARE, a contradiction. Now, having proved that the original survey does not contain asteroidal triples, the application of Theorem 6 in Tucker (1972) guarantees that the original survey must satisfy c1p.

We now show that given that the original survey satisfies c1p, it is rationalizable. Given the finiteness of $\mathcal{Q}$, $\succ$ is representable and, hence, there exists a collection of real values $\{\mu_q\}_{q \in \mathcal{Q}}$ such that, for every pair of questions $q, q' \in \mathcal{Q}$, $\mu_q > \mu_{q'}$ if and only if $q \succ q'$. For every $n \in \mathcal{N}$ such that $|Q_n| \geq 1$, consider any utility function $U_n(x)$ that is strictly increasing whenever $x \leq \frac{\min_{q \in Q_n} \mu_q + \max_{q \in Q_n} \mu_q}{2}$, and strictly decreasing otherwise. Moreover, we can select U_n to satisfy $U_n(\min_{q \in Q_n} \mu_q) = U_n(\max_{q \in Q_n} \mu_q) = 0$. This function is strictly quasi-concave. Set $\tau_n = 0$. Since Q_n is formed by consecutive

³²The recursive argument is needed only when $K \geq 3$; otherwise, the implication follows directly.

questions in $\succ$, it is immediate that $Q_n = \{q : U_n(\mu_q) \geq \tau_n = 0\}$, proving rationalization. For every $n \in \mathcal{N}$ with $|Q_n| = 0$, we can rationalize responses by considering U_n to be strictly negative and strictly quasi-concave, and setting $\tau_n = 0$.

We now prove that the extremeness relation directly revealed by the expanded D-survey coincides with the overlapping closure of the extremeness relation of the original D-survey, implying that (3) is equivalent to (2). Consider again any appropriate sequence of D-surveys $\{Q_n\}_{n \in \mathcal{N}_m}$, $m = 0, 1, \dots, M$, leading to the expanded D-survey. Denote by $|_m$ the extremeness revelations of survey $\{Q_n\}_{n \in \mathcal{N}_m}$. We need to show that $|_M = \widehat{|_0}$. Consider two successive D-surveys $\{Q_n\}_{n \in \mathcal{N}_m}$ and $\{Q_n\}_{n \in \mathcal{N}_{m+1}}$ in the sequence, and assume that $Q_i \cup Q_j$, with $Q_i, Q_j \in \{Q_n\}_{n \in \mathcal{N}_m}$, is the novel response in $\{Q_n\}_{n \in \mathcal{N}_{m+1}}$. We claim that $|_{m+1}$ consists of $|_m$ together with every relation obtained by overlapping $q |_m \{q', q_1\}$ and $q |_m \{q', q_2\}$ in which $q \notin Q_i \cup Q_j$, $q_1 \in Q_i \setminus Q_j$ and $q_2 \in Q_j \setminus Q_i$. First, it is obvious that any such derived relation will be revealed by $Q_i \cup Q_j$ and, hence, present in $|_{m+1}$. Second, consider any revelation $q |_{m+1} \{q', q''\}$ that was not present in $|_m$. It must be the case that this revelation is generated by response set $Q_i \cup Q_j$, and for it to be novel, it must be $q \notin Q_i \cup Q_j$, $q' \in Q_i \setminus Q_j$ and $q'' \in Q_j \setminus Q_i$. But in such a case, since Q_i and Q_j are overlapping sets, there must exist $q^* \in Q_i \cap Q_j$, and the D-survey $\{Q_n\}_{n \in \mathcal{N}_m}$ must reveal $q |_m \{q^*, q'\}$ (via Q_i) and $q |_m \{q^*, q''\}$ (via Q_j), and hence the novel revelation can be seen as the result of overlapping two original revelations. Using this recursively, we have shown that the extremeness revelations of the expanded D-survey can be seen as applications of overlapping to the original extremeness relation and hence $|_0 \subseteq |_M \subseteq \widehat{|_0}$. We now claim that $|_M$ is closed under overlapping. To see this, suppose that we observe $q |_M \{q^*, q'\}$ and $q |_M \{q^*, q''\}$. Then, we know that there exist Q_i, Q_j in the expanded D-survey such that $q^* \in Q_i \cap Q_j$ and $q' \in Q_i$ and $q'' \in Q_j$. If $q'' \in Q_i$ or $q' \in Q_j$, we can use the corresponding set to claim that $q |_M \{q', q''\}$. If neither is true, then Q_i and Q_j are overlapping sets and, since the expanded D-survey is closed under overlapping unions, $Q_i \cup Q_j$ belongs to the expanded D-survey, and we can use it to claim that $q |_M \{q', q''\}$. Since the overlapping closure is the minimal ternary relation containing $|$ and closed under overlapping, the following must hold $\widehat{|} \subseteq |_M$.

Finally, we establish the computational complexity of survey rationalizability, which involves two steps. The first computes the directly revealed extremeness relation, for which reading the data once suffices: for every individual n and every pair of questions in Q_n , every question outside Q_n is revealed extreme in the corresponding triple. There

are at most $N\binom{Q}{2}$ such pairs, each requiring $O(Q)$ operations, so this step has order $O(NQ^3)$. For the second, fix a question q and consider the graph G_q whose nodes are the remaining questions and in which two distinct questions q' and q'' are joined by an edge of weight 1 whenever $q \mid \{q', q''\}$, with every other pair of distinct nodes receiving weight ∞ . We denote these weights by $[q, \{q', q''\}, 0]$. Note that for every pair of distinct questions $q', q'' \neq q$, $q \widehat{\mid} \{q', q''\}$ if and only if q' and q'' are joined by a path in G_q : successive applications of overlapping along a path generate the relation between its endpoints, while an application of overlapping can never join questions belonging to different connected components. It therefore suffices to run the Floyd-Warshall algorithm on G_q . Denoting by $[q, \{q', q''\}, l]$ the weight of the shortest path between q' and q'' in G_q that uses only questions $\{1, \dots, l\}$ as intermediate nodes, these values satisfy

$$[q, \{q', q''\}, l] = \min \{ [q, \{q', q''\}, l-1], [q, \{q', l\}, l-1] + [q, \{q'', l\}, l-1] \},$$

with the conventions $[q, \{q', q'\}, l] = 0$ and $[q, \{q, q'\}, l] = \infty$ for $q' \neq q$. We iterate this process until l reaches Q . By the standard argument for the Floyd-Warshall algorithm, for every pair of distinct questions $q', q'' \neq q$, $[q, \{q', q''\}, Q]$ is finite if and only if q' and q'' are joined by a path in G_q , that is, if and only if $q \widehat{\mid} \{q', q''\}$. Running this procedure for each of the Q questions requires filling at most $Q\binom{Q}{2}Q$ entries, each in constant time, so this step has order $O(Q^4)$.³³ Once the overlapping closure has been computed, checking SARE requires inspecting at most $\binom{Q}{3}$ triples and hence $O(Q^3)$ further operations. Adding up, survey rationalizability is determined in $O(NQ^3 + Q^4)$ operations. This concludes the proof of the theorem. $\blacksquare$

Proof of Theorem 2: To see the only if part, suppose that P is rationalizable. Then, it must be so for some collection of parameters $[\{U_n, \{\tau_n^l\}_{l \in \mathcal{L}^*}\}_{n \in \mathcal{N}}, \{\mu_q\}_{q \in \mathcal{Q}}]$. Consider the D-survey $\{Q_{(n,l)}^P\}_{(n,l) \in \mathcal{N} \times \mathcal{L}^*}$. Associate the utility function U_n and the threshold τ_n^l with response set $Q_{(n,l)}^P$. The resulting D-survey is rationalizable. We can then use the only if part of Theorem 1 and the expanded D-survey $\{Q_n^P\}_{n \in \mathcal{N}_M}$ must satisfy WARE.

Suppose now that $\{Q_n^P\}_{n \in \mathcal{N}_M}$ satisfies WARE. By using the if part of Theorem 1, the associated D-survey $\{Q_{(n,l)}^P\}_{(n,l) \in \mathcal{N} \times \mathcal{L}^*}$ must satisfy c1p. Let $\succ$ be a linear order on $\mathcal{Q}$ under which c1p holds and consider any set of distinct values $\{\mu_q\}_{q \in \mathcal{Q}}$ representing this order. For every individual $n \in \mathcal{N}$, the sets $Q_{(n,l)}^P$ are nested intervals. Hence, when

³³Since all the edges of G_q have the same weight, only connectedness matters, so that the connected components of each G_q could be computed in $O(Q^2)$ operations, reducing this step to $O(Q^3)$.

the questions are ordered according to their locations, their labels weakly increase up to the block of questions receiving the highest label used by n and weakly decrease thereafter. Choose any question q_n^* in this block. Since $\mathcal{Q}$ is finite, we can assign utility values satisfying

$$P(n, q) < U_n(\mu_q) < P(n, q) + 1$$

for every $q \in \mathcal{Q}$, so that they strictly increase up to $\mu_{q_n^*}$ and strictly decrease thereafter. Piecewise linear interpolation through these values, together with strictly monotone extensions outside the range of question locations, yields a strictly quasi-concave function U_n . Set $\tau_n^l = l$ for every $l \in \mathcal{L}^*$. Then every question lies in exactly the utility band corresponding to its observed label. Rationalization follows. ■

Proof of Theorem 3: The proof is analogous to that of Theorem 2, so we only mention the most significant differences. Notice that from S we can construct a set of at most $Q \times N$ labels, and define the P-survey with the response map P having the property that, for every $n, n' \in \mathcal{N}$ and $q, q' \in \mathcal{Q}$, $P(n, q) \geq P(n', q')$ if and only if $S(n, q) \geq S(n', q')$. This allows us to construct the values μ_q and the functions U_n exactly as in Theorem 2. For every τ_n such that there exists $q \in \mathcal{Q}$ with $U_n(\mu_q) = \tau_n$, set $F_n(\tau_n) = S(n, q)$. Now, for every individual $n \in \mathcal{N}$, a larger threshold corresponds to a question with a larger utility and, given the construction and the fact that $P(n, q) \geq P(n, q')$ if and only if $S(n, q) \geq S(n, q')$, a larger utility corresponds to a weakly larger probability of endorsement. We can then extend these values to a CDF F_n over the reals, and rationalization follows. ■

Proof of Proposition 1: Let the survey be rationalizable with the locations of questions and the utility functions being $\{\mu_q\}_{q \in \mathcal{Q}}$ and $\{U_n\}_{n \in \mathcal{N}}$. Assume $\mu_Q > \mu_{Q-1} > \dots > \mu_1$. Hence, $Q \succ Q-1 \succ \dots \succ 1$. Recall from the proof of Theorem 1 that each of the responses in the expanded D-survey is formed by consecutive questions in $\succ$.

We now claim that for every question $i < Q-1$, the expanded D-survey must reveal $i \mid \{j, m\}$ for every pair of different questions $j, m > i$. By Richness, for every $i < j < Q$, there exists an individual $n_j \in \mathcal{N}_M$ such that $Q_{n_j} \cap \{j-1, j, j+1\} = \{j, j+1\}$. By rationalizability, since Q_{n_j} contains $j, j+1$ and excludes $j-1$, it must also exclude i . Notice now that the expanded D-survey is closed under unions of responses with

nonempty intersection.³⁴ Since Q_{n_j} and $Q_{n_{j+1}}$ both contain $j + 1$, and since the union already constructed at every step contains Q_{n_j} , it follows recursively that $\bigcup_{j=i+1}^{Q-1} Q_{n_j}$ belongs to the expanded D-survey. This response contains every question above i and excludes i , and hence reveals $i \mid \{j, m\}$ for every pair of different questions $j, m > i$. The argument for revealing the extremeness of i relative to any pair $j, m < i$ is symmetric. Therefore, for every triple of different questions satisfying $i > j > k$, the expanded D-survey reveals $i \mid \{j, k\}$ and $k \mid \{i, j\}$. Since every response in the expanded D-survey is an interval with respect to $\succ$, no further extremeness relation can be revealed and, by the proof of Theorem 1, the extremeness relation revealed by the expanded D-survey coincides with the indirect revealed relation of the original D-survey. Hence, all and only the extremeness relations of $\succ$ have been obtained through the indirect extremeness revelations.

Finally, to show uniqueness, consider any other order $\succ'$ different from $\succ$ and its inverse. Hence, there must be three questions $i > j > k$ where the intermediate one is different from j . Without loss of generality, let i be the intermediate question in $\succ'$. But then, we know that there is a response in the expanded D-survey such that $Q_m \cap \{i, j, k\} = \{j, k\}$, and this is not rationalizable for $\succ'$. If the expanded D-survey is not rationalizable with $\succ'$, the proof of Theorem 1 guarantees that the original survey cannot be rationalizable with questions located in a way compatible with $\succ'$, concluding the proof. ■

Proof of Proposition 2: Let the α -survey be rationalizable with the locations of questions and the utility functions being $\{\mu_q\}_{q \in \mathcal{Q}}$ and $\{U_n\}_{n \in \mathcal{N}}$. Given Proposition 1, every rationalization of the α -survey is compatible with the linear order $\succ$ implied by $\{\mu_q\}_{q \in \mathcal{Q}}$, up to inversion. Moreover, $\succ$ is common to the three survey formats. Fix $n \in \mathcal{N}$.

First, we show that in every rationalization the ideal view of n must belong to $\alpha(n)$. We prove this by way of contradiction. Suppose, for example, that it occupies a position equal to or above the question immediately above $\bar{q}_n^\alpha$. This question receives a strictly weaker response than $\bar{q}_n^\alpha$ and hence has strictly lower utility. But it lies between $\bar{q}_n^\alpha$ and the ideal view, possibly coinciding with the latter, contradicting strict quasi-concavity or maximality, respectively. The other case is symmetric.

³⁴Indeed, if two intersecting responses are nested, their union is one of them and hence already belongs to the expanded D-survey, while if they are not nested, they are overlapping and their union belongs to the expanded D-survey by construction.

Second, suppose that the three survey formats are related by assumption (ii). Under this assumption, the P-responses preserve the order of the S-responses and the D-responses preserve that of the P-responses. In particular, every question with the highest endorsement probability has the highest label and every question with the highest label is endorsed. The corresponding maximum-response sets are therefore nested, and $S(n) \subseteq P(n) \subseteq D(n)$.

Third, consider any location x in $\alpha(n)$. Hence, the response of n weakly increases below x and weakly decreases above it. We can therefore assign utility values that strictly increase up to x and strictly decrease thereafter, with stronger responses receiving higher utility. Piecewise linear interpolation and strictly monotone extensions yield a strictly quasi-concave function U_n uniquely maximized at x . A threshold rationalizes the D-survey, ordered thresholds rationalize the P-survey, and the assignments $F_n(U_n(\mu_q)) = S(n, q)$ can be extended to a CDF for the S-survey. Note that the above argument is independent of the rationalizations involving the other individuals in the survey, since $\succ$ is preserved. Thus every position in $\alpha(n)$ is attainable and no further identification involving μ_n is possible. This completes the proof. ■

Proof of Proposition 3: Consider an S-survey with exponential responses. By Proposition 1, we know that $\succ$ is fully identified, and we set it to be $1_q \succ 0_q$. Since the endorsement probability is strictly decreasing in the distance to μ_n , for every $n \in \mathcal{N}$, the interval $S(n)$ contains either one or two questions. As in the proof of Proposition 2, denote the highest and lowest questions in $S(n)$ by $\bar{q}_n^S$ and $\underline{q}_n^S$, respectively. Thus, either $\bar{q}_n^S \succ \underline{q}_n^S$ or $\bar{q}_n^S = \underline{q}_n^S \equiv q_n^S$. We now construct a map $h_n : \mathcal{Q} \rightarrow \{0, 1\}$ for each individual such that $h_n(q) = 1$ if and only if $\mu_q \geq \mu_n$ and $h_n(q) = 0$ otherwise. To do so, we consider two cases.

If $S(n)$ contains two questions, the exponential model of responses guarantees that μ_n must occupy a position strictly between these two questions, and we can assign $h_n(q) = 1$ to $\bar{q}_n^S$ and to every question above this one, according to $\succ$, and zero otherwise.

If $S(n)$ contains one question, the map can be defined by $h_n(q) = 1$ whenever $q \succ q_n^S$ and $h_n(q) = 0$ whenever $q_n^S \succ q$. However, the relative location of q_n^S is yet undetermined. To determine it, we consider another individual n' for which $h_{n'}(q_n^S)$ is known, and two questions q', q'' for which $h_n(q'), h_n(q''), h_{n'}(q'), h_{n'}(q'')$ are all known. Richness guarantees that such an individual and such questions exist. Indeed, since $Q \geq 4$, there are at least two questions on one side of q_n^S and, together with the

two closest of them, q_n^S forms a triple of consecutive questions in which it is extreme. Richness then provides a response containing these two questions and excluding q_n^S and, since every response in the expansion is a union of original responses, there is an individual n' and a question r with an original response $Q_{(n',r)}^S$ containing one of them, say q , and excluding q_n^S . Hence $S(n', q) \geq S(n', r) > S(n', q_n^S)$, and q_n^S is not a most frequently endorsed question of n' , so that $h_{n'}(q_n^S)$ is known. Finally, the side indicators of n and n' are known except possibly at their own most frequently endorsed questions, so it suffices to choose q' and q'' different from q_n^S and, if n' also has a single most frequently endorsed question, different from it as well; this is possible because $Q \geq 4$. It is immediate to see that the exponential model requires that

$$\begin{aligned} \frac{(-1)^{h_n(q_n^S)} \log S(n, q_n^S) - (-1)^{h_n(q'')} \log S(n, q'')}{(-1)^{h_n(q')} \log S(n, q') - (-1)^{h_n(q'')} \log S(n, q'')} &= \frac{\mu_{q_n^S} - \mu_{q''}}{\mu_{q'} - \mu_{q''}} = \\ &= \frac{(-1)^{h_{n'}(q_n^S)} \log S(n', q_n^S) - (-1)^{h_{n'}(q'')} \log S(n', q'')}{(-1)^{h_{n'}(q')} \log S(n', q') - (-1)^{h_{n'}(q'')} \log S(n', q'')}. \end{aligned}$$

Hence, $h_n(q_n^S)$ can be determined.³⁵

Given the complete map h_n , which is now available in both cases, notice that, for every q other than 0_q and 1_q we have

$$\begin{aligned} \frac{(-1)^{h_n(q)} \log S(n, q) - (-1)^{h_n(0_q)} \log S(n, 0_q)}{(-1)^{h_n(1_q)} \log S(n, 1_q) - (-1)^{h_n(0_q)} \log S(n, 0_q)} &= \mu_q, \\ \frac{-(-1)^{h_n(0_q)} \log S(n, 0_q)}{(-1)^{h_n(1_q)} \log S(n, 1_q) - (-1)^{h_n(0_q)} \log S(n, 0_q)} &= \mu_n. \end{aligned}$$

All locations are fully determined, concluding the proof. ■

Proof of Proposition 4: Consider the collection of D-survey responses $\{Q_n\}_{n \in \mathcal{N}}$. We construct a rationalization of these responses in a two-dimensional space. Locate all questions at different positions on the circumference of the unit circle centered at the origin. Consider Q_n . If $|Q_n| = 0$, rationalization is trivial by considering Euclidean preferences centered at the origin, with utility 0 at the origin and $\tau_n = 0$. If $|Q_n| = 1$, rationalization is trivial, by considering Euclidean preferences centered at the only question endorsed, with utility 0 at that point, and $\tau_n = 0$. If $|Q_n| \geq 3$, consider the convex hull of the locations of the questions in Q_n ; denote this set by $CH(Q_n)$. Given that all questions in $\mathcal{Q}$ are at distinct points on the unit circle, it is evident that

³⁵Notice that in the extreme case in which $S(n, q_n^S) = 1$, it must be $\mu_n = \mu_{q_n^S}$ and hence $h_n(q_n^S) = 1$. Its value is immaterial for the displayed formulas, since $\log S(n, q_n^S) = 0$.

$q \in CH(Q_n) \cap \mathcal{Q}$ if and only if $q \in Q_n$; every question in $\mathcal{Q} \setminus Q_n$ lies outside this convex hull. We construct the utility function U_n as follows. Select any point in the interior of $CH(Q_n)$ and denote it by μ_n . Then,

- $U_n(\mu_n) = 0$.
- For every y on the boundary of $CH(Q_n)$, set $U_n(y_1, y_2) = -1$.
- For every x outside $CH(Q_n)$, let y be the unique point on the boundary of $CH(Q_n)$ such that $y = \lambda x + (1 - \lambda)\mu_n$. We can then set $U_n(x) = \frac{-1}{\lambda}$.
- For every x in the interior of $CH(Q_n)$, let y be the unique point on the boundary of $CH(Q_n)$ such that $x = \lambda y + (1 - \lambda)\mu_n$. We can then set $U_n(x) = -\lambda$.

By construction, all indifference curves have the same shape (expansions or contractions of the polygon formed by the locations of the questions positively endorsed by the individual), with utility normalized to 0 and -1 for the ideal view and the polygon indifference curve. The resulting function is concave, assigns utility -1 to every endorsed question, and assigns strictly lower utility to every nonendorsed question. Given the finiteness of the survey, for sufficiently small $\varepsilon > 0$, replacing $U_n(x)$ with $\tilde{U}_n(x) = U_n(x) - \varepsilon \|x - \mu_n\|^2$ preserves this separation and yields a strictly concave, and hence strictly quasi-concave, utility function. Choosing τ_n between the utility values of the endorsed and nonendorsed questions, or below all endorsed questions when $Q_n = \mathcal{Q}$, rationalizes the responses of the individual.

If $|Q_n| = 2$, select a third point on the circumference that does not correspond to the location of any question, and consider the triangle formed by these three points. Then, proceed to construct the utility function and the threshold as per the above reasoning. This again rationalizes the set of responses in a two-dimensional space, concluding the proof. $\square$

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